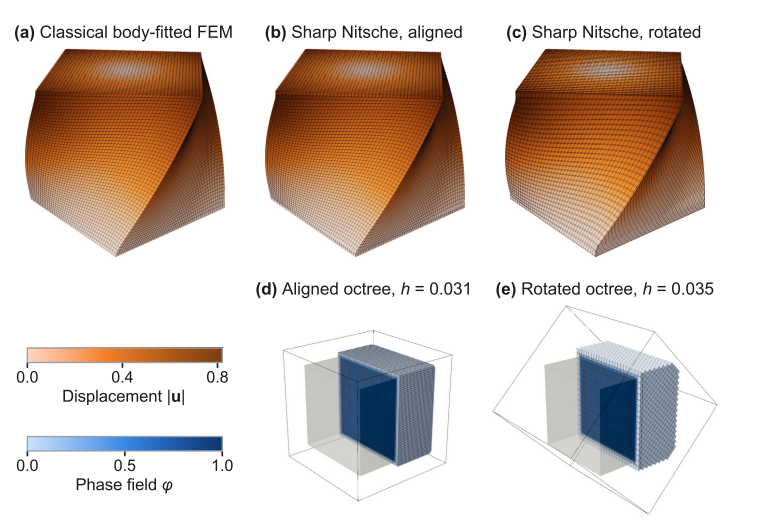

(a) Classical body-fitted FEM
(b) Sharp Nitsche, aligned
(c) Sharp Nitsche, rotated
(d) Aligned octree, h = 0.031
(e) Rotated octree, h = 0.035
0.0
0.4
0.8
Displacement |u|
0.0
0.5
1.0
Phase field φ
(f) Convergence to the body-fitted reference
Relative L2 error in u
10−1
10−2
10−3
(g) Cost of 20 load steps
Solve time (s)
Newton iterations
40–74 sharp-volume Nitsche
74–100 body-fitted
10^1
10^0
1/4
1/8
1/16
1/32
Element size h (box size 1.5×1.5×1.5)
Classical body-fitted FEM
Sharp-volume Nitsche, aligned
Sharp-volume Nitsche, rotated